Estimating peer effects in noisy, low-rank networks via network smoothing

Alex Hayes and Keith Levin

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Oregon State University

Roadmap for today

1. What are peer effects and how do we estimate them?
2. Peer effects in latent spaces
3. Estimating peer effects from noisy data
4. Shameless self-promotion

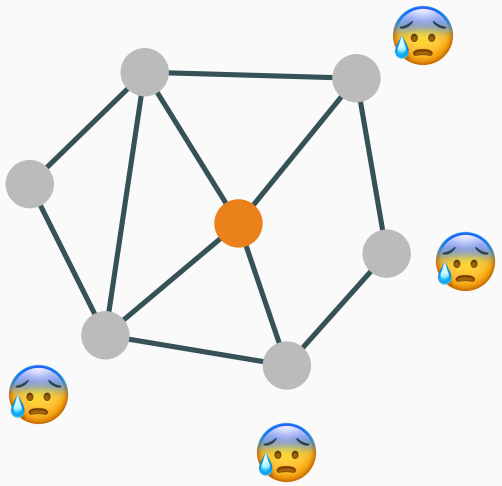
Peer effects

Humans constantly influence their social contacts



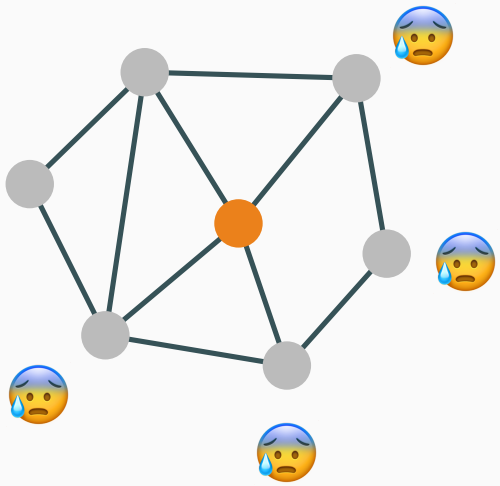
Data on 241 MBA students from “Do Your Friends Stress You out? A Field Study of the Spread of Stress…” by Li et al. (2023)

These social influences are called “peer effects” or “spillovers”



Contagion: if my friends are anxious 😰, I am more likely to be anxious 😰

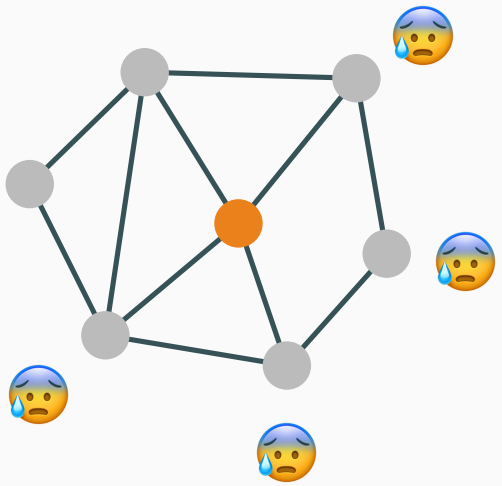
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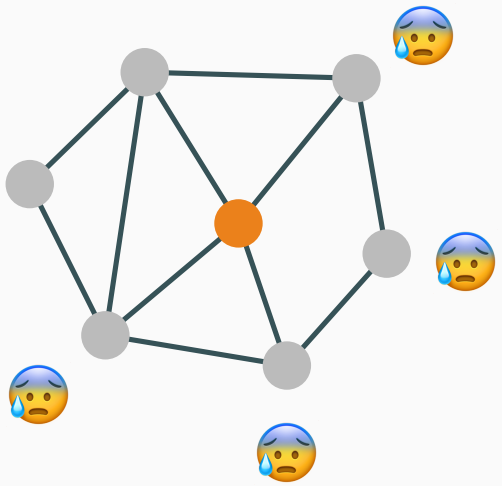
* Can be defined counterfactually, but we do not consider counterfactual inference in this talk.

Measuring someone's friends is challenging



If I ask you who your friends are:

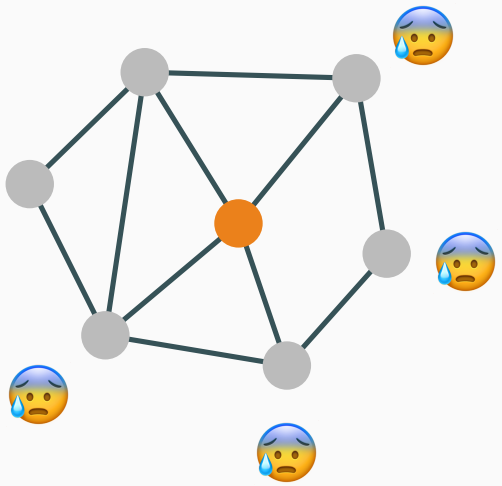
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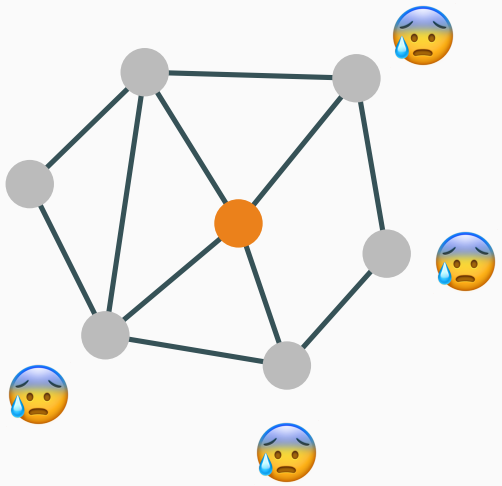
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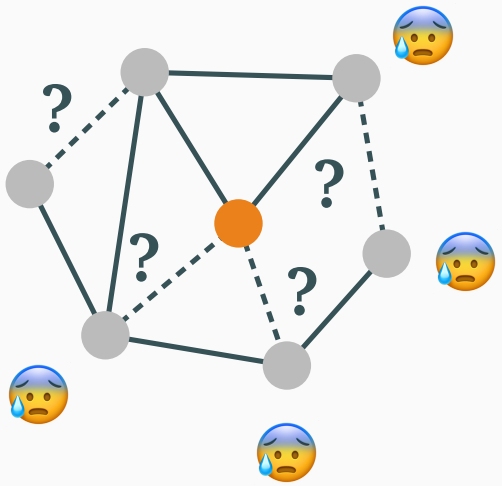
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Measuring someone's friends is challenging



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- answers vary depending on how exactly I ask you,
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- you might think someone is your friend, but they might disagree.

Peer effects estimators need perfectly observed networks*

$$\underbrace{Y_i}_{\text{outcome}} = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{W_i\beta_w}_{\text{covariate effects}} + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{average outcome among friends}} + \underbrace{\epsilon_i}_{\text{noise}}$$

$A_{ij} \in \{0, 1\}$ indicates a friendship between i and j, and $d_i = \sum_j A_{ij}$ is i's number of friends

* See Griffith and Kim (2024), Griffith (2022), Lewbel et al. (2024), Lewbel et al. (2025), Li et al. (2026) and Boucher and Houndetoungan (2025).

Hold on! Does this model even make sense?

Write the model in matrix form, with $G_{ij} = A_{ij}/d_i$ the degree-normalized adjacency matrix:

$$Y = \beta_0 1_n + W\beta_w + \beta_y GY + \epsilon$$

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$$\overset{*}{=} \underbrace{\sum_{k=0}^{\infty} \beta_y^k G^k}_{\substack{\text{repeated} \\ \text{neighborhood} \\ \text{averaging}}} \left(\beta_0 1_n + W\beta_w + \epsilon\right)$$

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* Must have $\left|\beta_y\right| < 1$, so the effect of averaging decays with repetition

Because GY and ε are correlated, cannot use ordinary least squares

Estimate via two-stage least squares with social instruments

$$X = [1_n \; W \; GY]$$ usual design matrix

$$H = [1_n \; W \; GW \; G^2W]$$ peer-of-peer instruments

$$M = H\left(H^\top H\right)^{-1} H^\top$$ projection onto instruments

Two-stage least squares estimator

$$\hat{\beta} = \left(X^\top M X\right)^{-1} X^\top M Y$$

Theorem (Kelejian and Prucha 1998): Under regularity conditions

$$\sqrt{n}\left(\hat{\beta} - \beta\right) \to \mathcal{N}\left(0, \Sigma_\beta\right)$$

Peer effects estimates need perfectly observed networks

$$\underbrace{Y_i}_{\text{outcome}} = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{W_i \beta_w}_{\substack{\text{covariate} \\ \text{effects}}} + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\substack{\text{average outcome} \\ \text{among friends}}} + \underbrace{\epsilon_i}_{\text{noise}}$$

$A_{ij} \in \{0, 1\}$ indicates a friendship between i and j, and $d_i = \sum_j A_{ij}$ is i's number of friends

…but the edges A_{ij} are always measured with noise!

Peer effects in latent spaces

We propose a model based on probability of friendship instead

$$Y_i = \beta_0 + W_i \beta_w + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{average outcome among friends}} + \epsilon_i \qquad \text{requires all } A_{ij}$$

$$Y_i = \theta_0 + W_i \theta_w + \underbrace{\theta_y \sum_{j \neq i} \frac{\mathbb{E}[A_{ij}]}{\tilde{d}_i} Y_j}_{\substack{\text{average outcome} \\ \text{weighted by} \\ \text{friendship probability}}} + \epsilon_i \qquad \text{requires } \mathbb{E}[A_{ij}]$$

where $d_i = \sum_j A_{ij}$ is degree and $\tilde{d}_i = \sum_j \mathbb{E}[A_{ij}]$ is expected degree.

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where $d_i = \sum_j A_{ij}$ is degree and $\tilde{d}_i = \sum_j \mathbb{E}[A_{ij}]$ is expected degree.

Questions: (1) How to get the expected friendships $\mathbb{E}[A_{ij}]$?, and (2) Is this a good idea?

We estimate friendships based on a low-rank network model



Intuition: stochastic blockmodels

Block indicators Z_i

Popularity parameters ω_i

Mixing matrix $B \in [0,1]^{r \times r}$

$$\mathbb{P}\left(A_{ij} = 1 \mid Z, \omega, B\right) = \underbrace{\omega_i Z_i B Z_j^\top \omega_j}_{\text{rank } r}$$

Only assume undirected, possibly weighted $A \in R^{n \times n}$ with low rank $P = \mathbb{E}[A \mid U, S] = USU^\top$ and sub-gamma (Bernoulli, Poisson, Gaussian, Gamma, …) deviations of A around P.

We estimate friendships based on a low-rank network model



Let $\hat{P} = \hat{U}\hat{S}\hat{U}^\top$ be rank-r SVD of A.

Then $\hat{P}_{ij} \to \mathbb{E}\left[A_{ij}\right]$

Principal components of A

Is the latent contagion model a good working model?

$$Y_i = \beta_0 + W_i \beta_w + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{average outcome among friends}} + \epsilon_i \qquad \text{requires all } A_{ij}$$

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where $d_i = \sum_j A_{ij}$ is degree and $\tilde{d}_i = \sum_j \mathbb{E}[A_{ij}]$ is expected degree.

In dense networks the design matrices should be close

The difference in the design matrices is

$$\underbrace{\sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{average outcome among friends}} - \underbrace{\sum_{j \neq i} \frac{\mathbb{E}[A_{ij}]}{\tilde{d}_i} Y_j}_{\text{average outcome weighted by friendship probability}} = \sum_{j \neq i} \left(\frac{A_{ij}}{d_i} - \frac{\mathbb{E}[A_{ij}]}{\tilde{d}_i} \right) Y_j$$

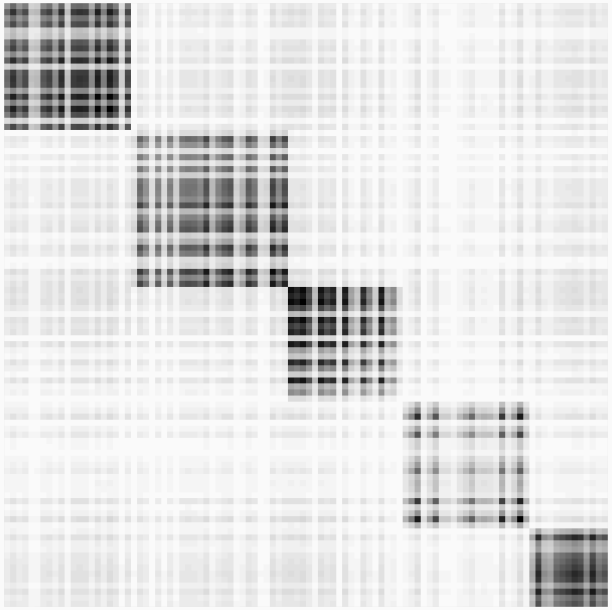
$$= \mathcal{O}_p\left(\frac{\sqrt{n}\log(n)}{d_i} \right)$$

where $A_{ij} - \mathbb{E}[A_{ij}]$ are i.i.d. Close when $d_i \gg \sqrt{n}\log(n)$!

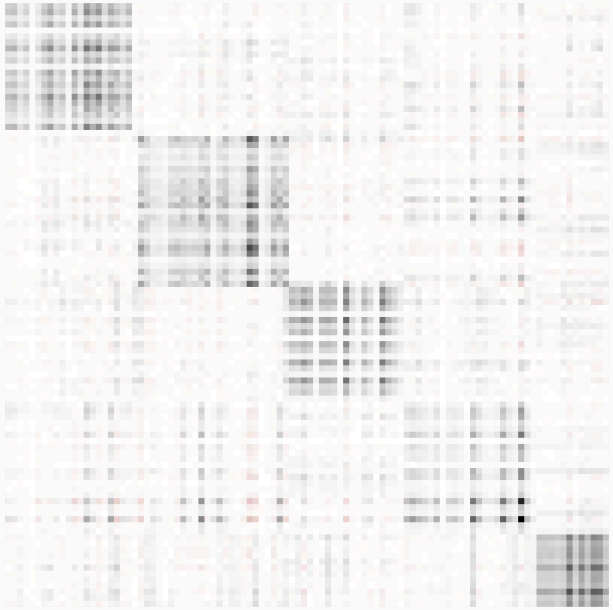
Why are latent contagion and peer contagion models so close?



A



$\mathbb{E}[A] = P$



$\hat{P}$

Estimate latent contagion model by plugging $\hat{P}$ in for A

Instead of using $G = D^{-1}A$, define $\hat{D} = \text{diag}\left(\hat{P}1_n\right)$ and $\hat{G} = \hat{D}^{-1}\hat{P}$.

peer contagion estimator

$$X = [1_n \; W \; GY]$$

$$H = \left[1_n \; W \; GW \; G^2W\right]$$

$$M = H\left(H^\top H\right)^{-1}H^\top$$

latent contagion estimator

$$\hat{X} = \left[1_n \; W \; \hat{G}Y\right]$$

$$\hat{H} = \left[1_n \; W \; \hat{G}W \; \hat{G}^2W\right]$$

$$\hat{M} = \hat{H}\left(\hat{H}^\top\hat{H}\right)^{-1}\hat{H}^\top$$

Latent contagion two-stage least squares estimator

$$\hat{\boldsymbol{\theta}} = \left(\hat{X}^\top\hat{M}\hat{X}\right)^{-1}\hat{X}^\top\hat{M}Y$$

Latent contagion is a reasonable working model

In dense networks,

Theorem: $\hat{\theta}$ is asymptotically normal under the latent contagion model

$$\sqrt{n}\left(\hat{\theta} - \theta\right) \to \mathcal{N}\left(0, \Sigma_{\theta}\right)$$

and also

Theorem: $\hat{\theta}$ is asymptotically normal under the peer contagion model

$$\sqrt{n}\left(\hat{\theta} - \beta\right) \to \mathcal{N}\left(0, \Sigma_{\beta}\right)$$

* Omitting some details about identification for intuition

What do β and θ mean when models are misspecified?

Let $\tilde{G} = \tilde{D}^{-1}P$ where $\tilde{D} = \text{diag}\left(\tilde{d}_1, ..., \tilde{d}_n\right)$.

Write

$$\mathbb{E}_A[\cdot] = \mathbb{E}[\cdot \mid A, W]$$
$$\mathbb{E}_P[\cdot] = \mathbb{E}[\cdot \mid P, W]$$

Then:

$$\mathbb{E}_A[X] = \mathbb{E}_A[1_n \; W \; GY], \quad M = H\left(H^\top H\right)^{-1} H^\top \qquad \text{peer contagion}$$

$$\mathbb{E}_P\left[\tilde{X}\right] = \mathbb{E}_P\left[1_n \; W \; \tilde{G}Y\right], \quad \widetilde{M} = \widetilde{H}\left(\widetilde{H}^\top \widetilde{H}\right)^{-1} \widetilde{H}^\top \qquad \text{latent contagion}$$

What do β and θ mean when models are misspecified?

Define the **population TSLS projection estimands**:

$$\beta^* = \left(\mathbb{E}_A[X]^\top M \mathbb{E}_A[X]\right)^{-1} \mathbb{E}_A[X]^\top M \mathbb{E}_A[Y]$$

$$\theta^* = \left(\mathbb{E}_P\left[\tilde{X}\right]^\top \tilde{M} \mathbb{E}_P\left[\tilde{X}\right]\right)^{-1} \mathbb{E}_P\left[\tilde{X}\right]^\top \tilde{M} \mathbb{E}_P[Y]$$

- Under peer contagion, $\beta^* = \beta$ and θ^* is the misspecified 2SLS target
- Under latent contagion, $\theta^* = \theta$ and β^* is the misspecified 2SLS target

Asymptotically, β and θ have the same interpretation

Theorem: In sufficiently dense low-rank networks, the population TSLS projection functionals are asymptotically equivalent:

$$\left\| \underbrace{\left(\mathbb{E}_A[X]^\top M \mathbb{E}_A[X]\right)^{-1} \mathbb{E}_A[X]^\top M \mathbb{E}_A[Y]}_{\beta^* \text{ (peer target)}} - \underbrace{\left(\mathbb{E}_P\left[\tilde{X}\right]^\top \widetilde{M} \mathbb{E}_P\left[\tilde{X}\right]\right)^{-1} \mathbb{E}_P\left[\tilde{X}\right]^\top \widetilde{M} \mathbb{E}_P[Y]}_{\theta^* \text{ (latent target)}} \right\| = o_p\left(\frac{1}{\sqrt{n}}\right)$$

If peer contagion is true ($\beta^* = \beta$), the latent target θ^* is within estimation error of β!

If latent contagion is true ($\theta^* = \theta$), the peer target β^* is within estimation error of θ!

Estimating peer effects from noisy data

The latent contagion estimator doesn't need A, only $\mathbb{E}(A)$

Recall $\hat{P} = \hat{U}\hat{S}\hat{U}^\top$ estimates $\mathbb{E}[A]$ and $\hat{D} = \text{diag}\left(\hat{P}1_n\right)$ and $\hat{G} = \hat{D}^{-1}\hat{P}$.

$$\hat{X} = \left[1_n \; W \; \hat{G}Y\right]$$

$$\hat{H} = \left[1_n \; W \; \hat{G}W \; \hat{G}^2W\right]$$

$$\hat{M} = \hat{H}\left(\hat{H}^\top\hat{H}\right)^{-1}\hat{H}^\top$$

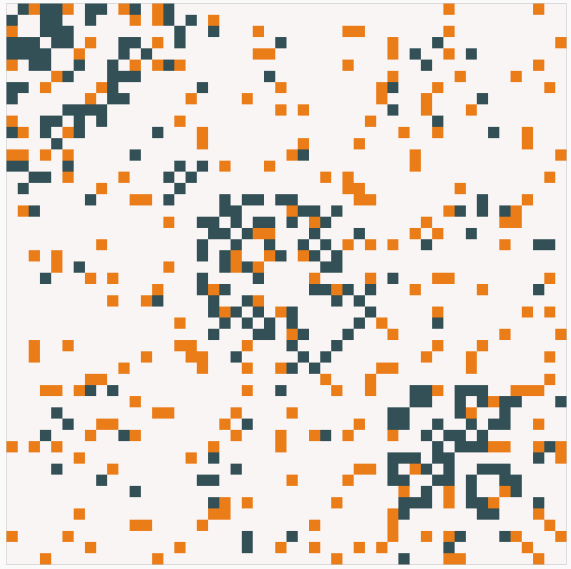
Latent contagion two-stage least squares estimator

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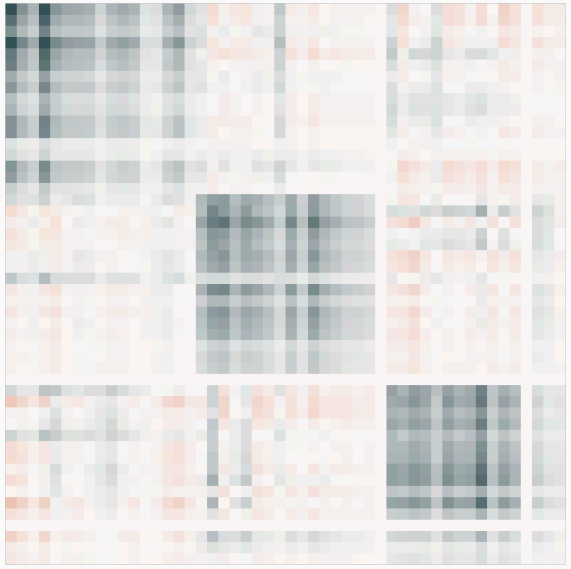
Only needs Y, W and an estimate $\hat{P}$ of low-rank P – use matrix completion!

Estimating friendship probabilities: partial network

Observed data



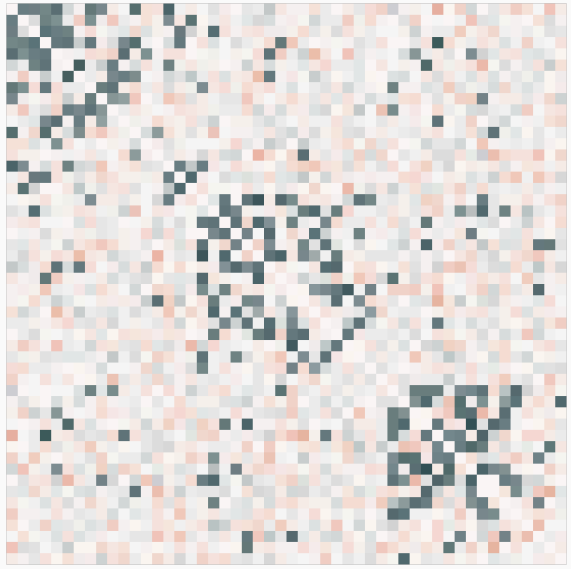
Estimated $\hat{P}$



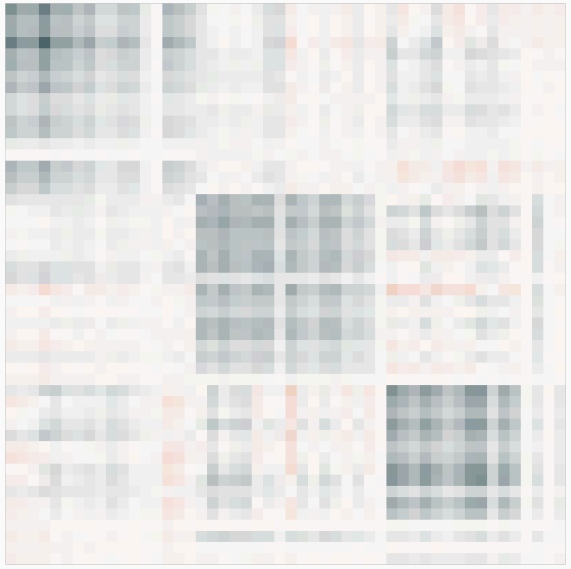
10% of edge missing at random (softImpute)

Estimating friendship probabilities: contaminated edges

Observed data



Estimated $\hat{P}$



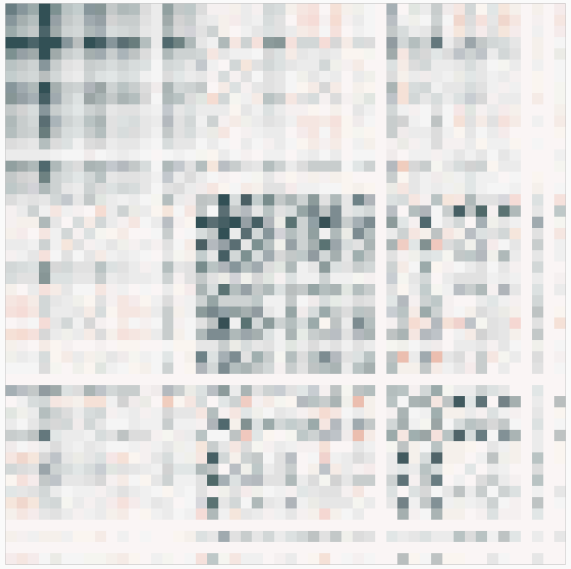
Edges contaminated by mean-zero i.i.d. noise (SVD)

Estimating friendship probabilities: ego-centric sampling

Observed data



Estimated $\hat{P}$



Only see neighborhoods for 50% of nodes in network (Chan and Li 2023)

Even more exotic: Aggregated Relational Data (ARD)

Traditional network measurement

- Who are you friends with?
- Who do you talk to often?

Aggregated Relational Data

How many people do you know who…

- study statistics?
- live in Portland?

Estimating friendship probabilities: aggregated relational data

Observed data

C (ARD counts)

1	1	0
1	1	0
2	0	1
0	1	1
0	0	1

$\mathbb{N}^{n \times k}$

$=$

A (network)

0	1	1	0	0
1	0	1	0	0
1	1	0	1	0
0	0	1	0	1
0	0	0	1	0

$\{0,1\}^{n \times n}$

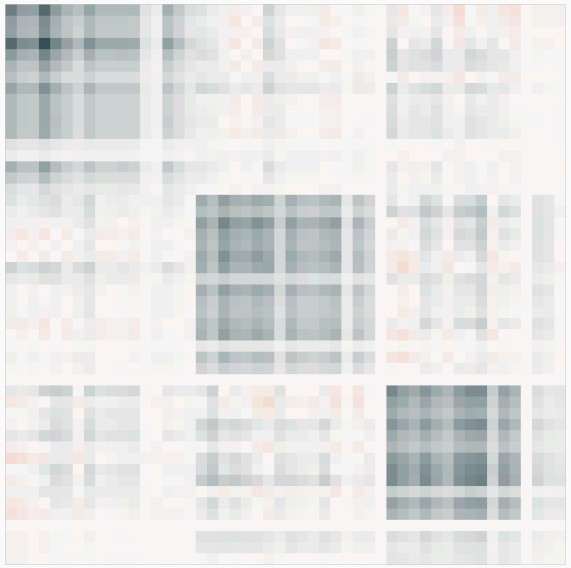
$\times$

W (traits)

1	0	0
1	0	0
0	1	0
0	0	1
0	0	1

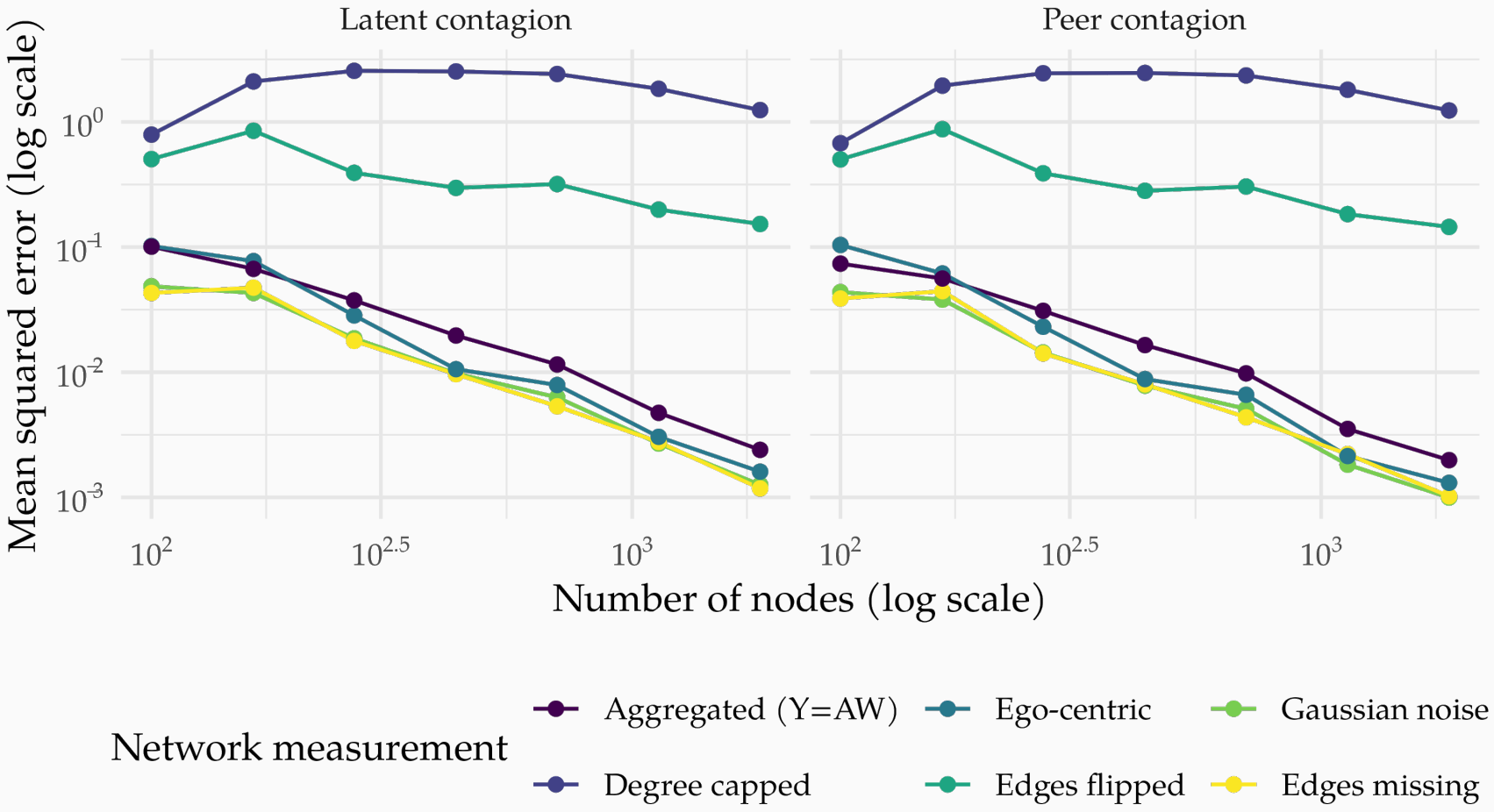
$\{0,1\}^{n \times k}$

Estimated $\hat{P}$



Only see friend trait counts and traits (Hayes et al 2026+)

Simulations validate our estimator with noisy data



Takeaway 1

Peer effects can be estimated without observing the network exactly: use estimated friendship probabilities as the "network."

Takeaway 2

We can use the entire toolbox of matrix completion techniques to handle network noise in low-rank networks.

Does plugging in $\hat{\mathbb{E}}[A]$ in for A work as a general technique to handle noise in networks?

Practical limitations of the latent contagion model

Density

Our theory requires moderately dense networks but most real world social networks are sparse

Identification

It's often hard to distinguish between contagion and localized outcomes in a social network (Hayes and Levin 2026; Shalizi and Thomas 2011)

Anxiety before final exams: specification matters

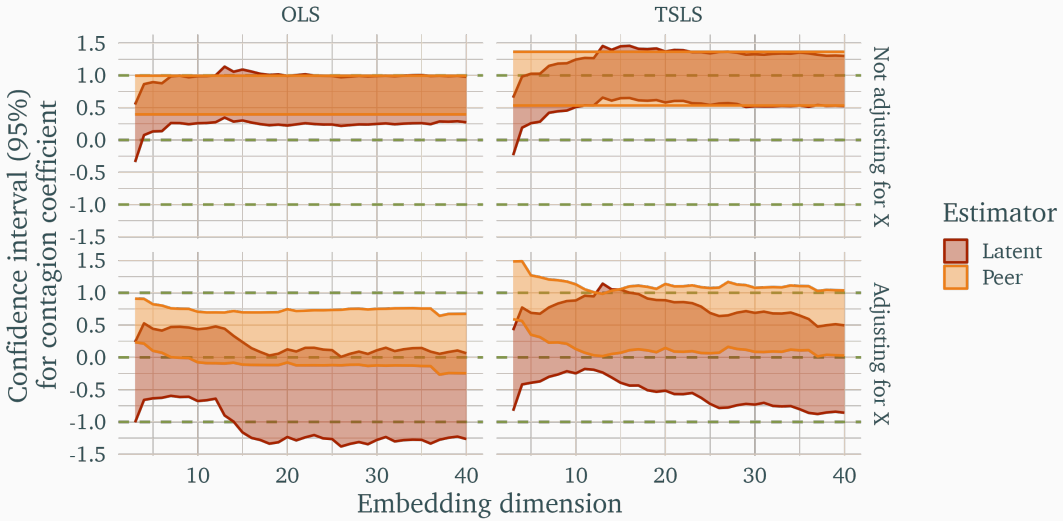
Data on 241 MBA students from "Do Your Friends Stress You out? A Field Study of the Spread of Stress…" by Li et al. (2023)

$$\text{anxiety} \sim \text{personality} + \text{demographics} + U + \text{latent peer anxiety}$$

$$\text{anxiety} \sim \text{personality} + \text{demographics} + U + \text{peer anxiety}$$

U accounts for localized/group effects.

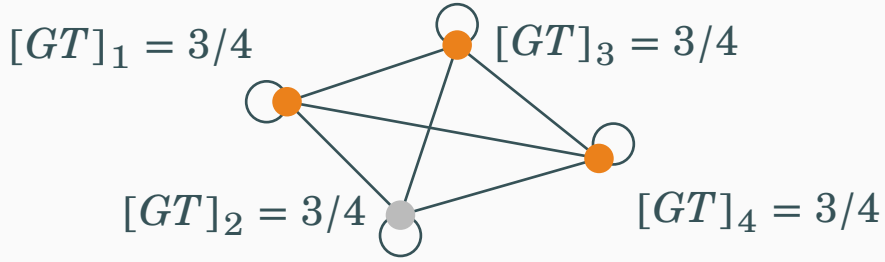
Dimension of U is unknown!



Model specification dramatically impacts inference

Shameless self-promotion

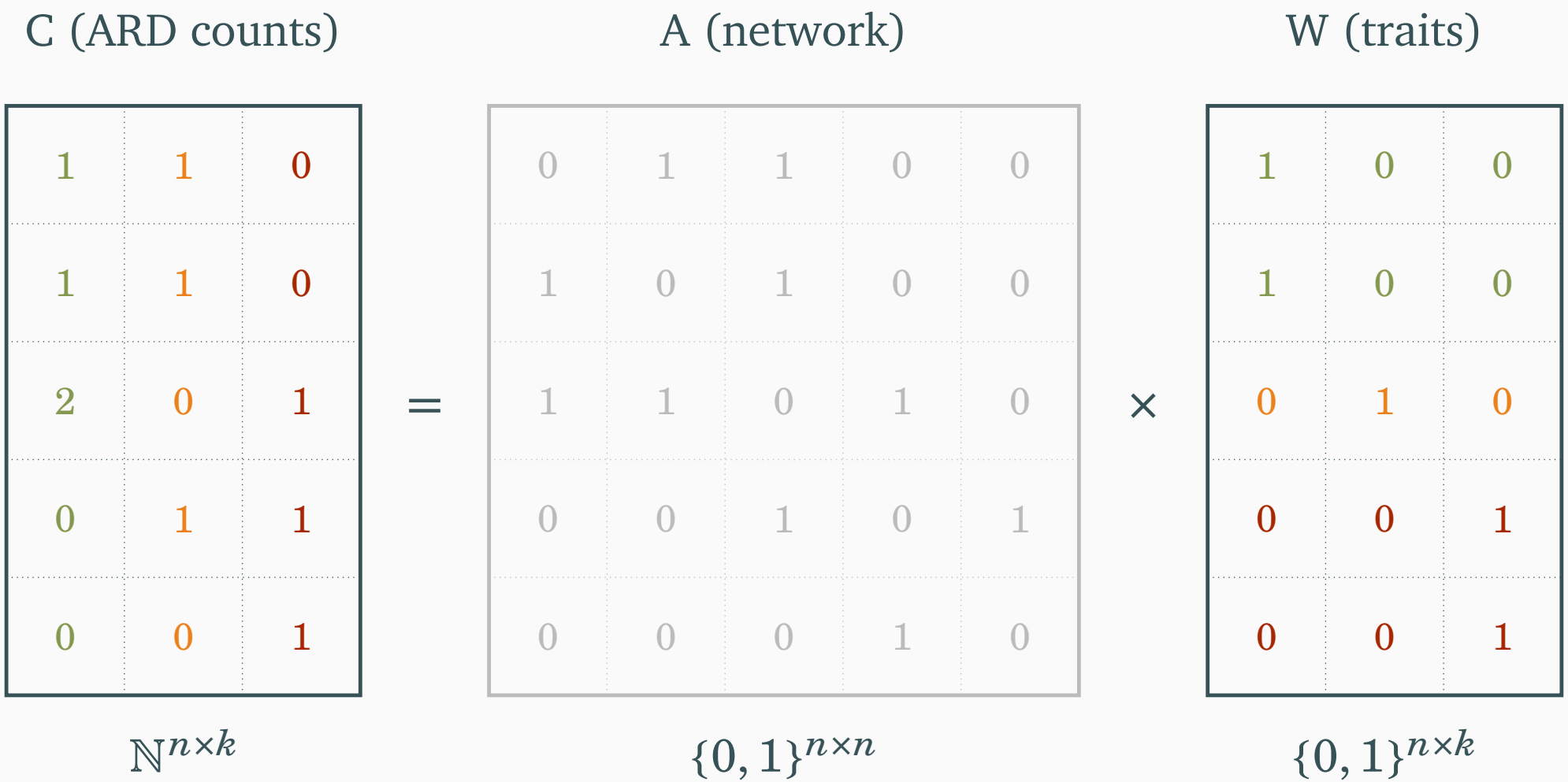
I’m currently working on: minimax rates for peer effects



$$\text{Col}(GT) \propto 1_n \implies \text{peer covariates collapse to the intercept!}$$

Signal-to-noise ratio can tend to zero in network regression models

I’m currently working on: Aggregated Relational Data



I'm currently working on: network central limit theorems

GENERAL COVARIANCE-BASED CONDITIONS FOR CENTRAL LIMIT THEOREMS WITH DEPENDENT TRIANGULAR ARRAYS

ARUN G. CHANDRASEKHAR[‡,§], MATTHEW O. JACKSON[‡,⋆], TYLER H. MCCORMICK[◇,¶], AND VYDHOURIE THIYAGESWARAN[◇]

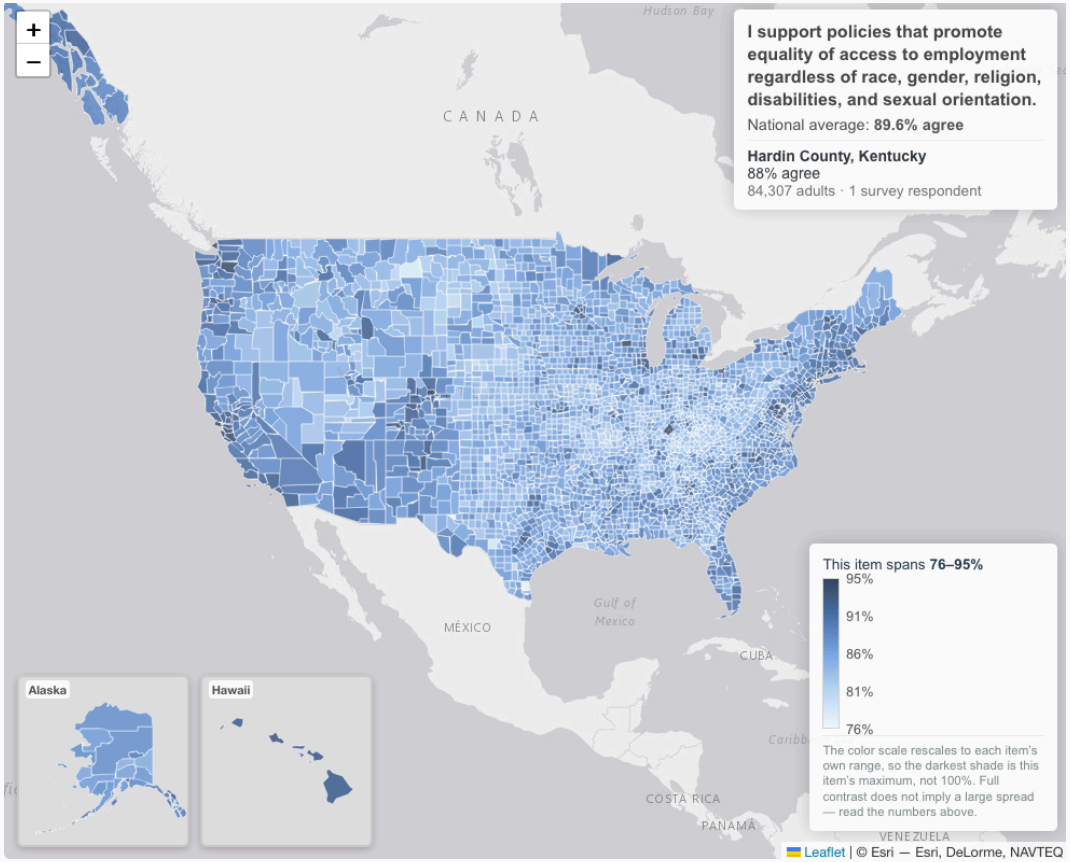
Abstract. We present a general central limit theorem with simple, easy-to-check covariance-based sufficient conditions for triangular arrays of random vectors when all variables could be interdependent. The result is constructed from Stein's method, but the conditions are distinct from related work. We show that these covariance conditions nest standard assumptions studied in the literature such as M-dependence, mixing random fields, non-mixing autoregressive processes, and dependency graphs, which themselves need not imply each other. This permits researchers to work with high-level but intuitive conditions based on overall correlation instead of more complicated and restrictive conditions such as strong mixing in random fields that may not have any obvious micro-foundation. As examples of the implications, we show how the theorem implies asymptotic normality in estimating: treatment effects with spillovers in more settings than previously admitted, covariance matrices, processes with global dependencies such as epidemic spread and information diffusion, and spatial processes with Matérn dependencies.

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1. Introduction

In many contexts researchers use an interdependent set of random vectors to develop estimators and need to study whether the estimators are asymptotically normal. Often, dependency is modeled in idiosyncratic ways, with perhaps unintuitive if not unappealing conditions to describe assumptions on correlation. It is not clear, for instance, if the α- or ϕ-mixing random field structures appropriately capture rainfall shocks in an empirical analysis, though the researcher is willing to assume that there is decaying covariance in some sense. Further, in many settings—such as treatment effects with spillovers, epidemic spread, information diffusion, general equilibrium effects, and so on,—the correlation is non-zero for any finite set of observations across all random vectors of interest, which is not allowed for in the literature.

I’m currently working on: maps of public opinion about DEI



Preliminary results: https://alexpghayes.github.io/wisdom

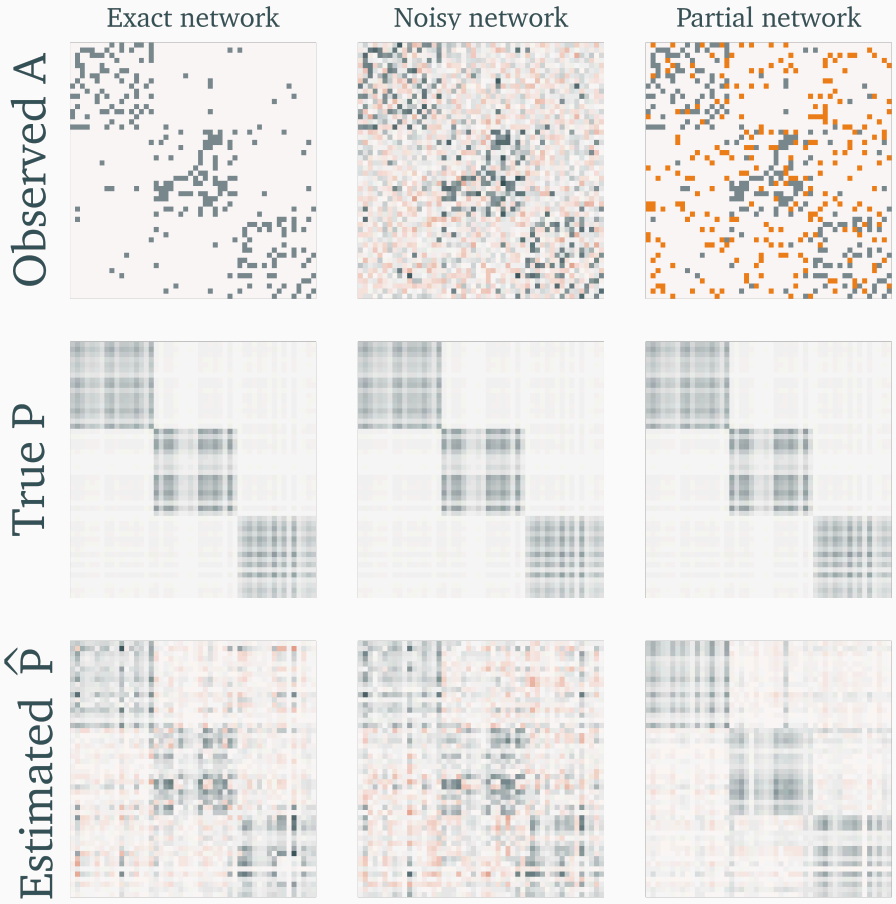
Thank you! Questions? Comments?

We propose one of the first methods to estimate peer effects that works with noisy friendship measurements

I am looking for **students** and **collaborators**

alex.hayes@oregonstate.edu

alexpghayes.com



Appendix

Assumptions of the canonical two-stage least squares estimator

1. All diagonal elements of G are zero (for $\hat{G}$, we set the diagonal of $\hat{P}$ to zero before row-normalizing)
2. $(I - \beta G)^{-1}$ is non-singular with $|\beta| < 1$.
3. G and $I - \beta G$ are uniformly bounded in row and column sums.
4. The regressor matrices W have full column rank (for n large enough), and the elements of W are uniformly bounded in absolute value.
5. ϵ_i are i.i.d. with zero mean and bounded variance $\mathbb{E}\left[\epsilon_i^2\right] = \sigma_\epsilon^2 < b < \infty$. Additionally ϵ_i have finite fourth order moments.

Assumptions of the canonical two-stage least squares estimator

Further, the instruments H have full column rank and are composed of a subset of linearly independent columns of $(1_n \;\; W \;\; GW \;\; G^2W)$ where the subset contains $[1_n \;\; W]$. Additionally

$$\lim_{n \to \infty} \frac{1}{n} H^\top H$$

is finite and non-singular and

$$\frac{1}{n} H^\top X$$

converges in probability to a matrix that is finite and has full column rank.

Assumptions of the canonical two-stage least squares estimator

If all the previous assumptions hold, then

$$\sqrt{n}\left(\hat{\boldsymbol{\beta}}_{\text{TSLS}} - \boldsymbol{\beta}\right) \to \mathcal{N}\left(0, \sigma_\epsilon^2 \operatorname{plim}\left(\left(\frac{1}{n} X^\top M X\right)^{-1}\right)\right)$$

recalling that

$$X = [1_n \; W \; GY] \qquad\qquad G = D^{-1}A$$

is the covariate matrix and

$$M = H\left(H^\top H\right)^{-1} H^\top$$

is the projection onto the instruments $H = \left[1_n \; W \; GW \; G^2 W\right]$.

Estimator

`ivreg(y ~ Gy + W | W + GW + G^2W)`, with instruments 1_n, W, GW, G^2W

Random dot product graphs

Definition. Let F be a distribution on R^r such that $0 \leq x^\top y$ for all $x, y \in \operatorname{supp} F$ and the convex cone of $\operatorname{supp} F$ is r-dimensional. Draw $Z_1, Z_2, \ldots, Z_n \overset{\text{iid}}{\sim} F$, and collect these in the rows of $Z \in R^{n \times r}$ for ease of notation. Conditional on these n vectors, which we call latent positions, generate edges by drawing $\{A_{ij} : 1 \leq i < j \leq n\}$ as independent (ν, b)-subgamma random variables with $\mathbb{E}\left[A_{ij} \mid Z\right] = \rho Z_i^\top Z_j$, where $\rho \in [0, 1]$. Then we say that A is distributed according to an n-vertex random dot product graph with latent position distribution F, (ν, b)-subgamma edges and sparsity factor ρ. We write $(A, Z) \sim \text{RDPG}(F, n)$.

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